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Kinetic modelling of Maillard reaction products and protein content during roasting of coffee beans

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ABSTRACT

In this study, the effects of various roasting methods on the α–dicarbonyl compounds (α-DCs), melanoidins, crude proteins, and colour value of roasted coffee beans (*Coffea arabica*) were analysed. The concentration of α-DCs in the coffee samples significantly changed ($p < 0.05$) during roasting, while that of melanoidins increased with roasting time in all roasting methods regardless of temperature. Crude protein decreased as roasting time increased. Through kinetic modelling, pan roasting was found to increase the rate constant of α-DCs, melanoidins, and crude proteins by up to 62.6%, 9.3%, and 33.3%, respectively. Air frying was observed to decrease the activation energy of α-DCs, melanoidins, and crude proteins by up to 57.8%, 22.3%, and 12.6%, respectively. The pan roasting method was shown to minimise α-DCs by up to 15% during medium roasting of coffee beans.

1. Introduction

Coffee, one of the most consumed beverages globally, commands a market valuation of 1449 billion US dollars (aTFIS, 2019). The two common species of green coffee, *Coffea arabica* and *Coffea robusta*, contribute approximately 70% and 30% to the overall global coffee market, respectively (Cagliani, Pellegrino, Giugno, & Consonni, 2013). Protein content ranges from 12.5% to 13.4% and polysaccharides from 38% to 55% in green coffee beans (Fischer, Reimann, Trovato, & Redgwell, 2001). During roasting, raw coffee beans undergo chemical reactions, such as the Maillard reaction (Wang, 2019; Kim, Choi, Lee, & Lee, 2021; Park, Noh, Kim, & Lee, 2024; Park, Jo, & Lee, 2021).

The Maillard reaction comprises three stages. In the initial stage, a Schiff base is produced as amino acids or amines readily react with the carbonyl functional group of a reducing carbohydrate, such as glucose (Kwon, Ahn, & Lee, 2021; Yu, Chu, Park, Park, & Lee, 2021). Through the 1,2-eneaminols formed from 1,2-dienediol, a rearrangement of the Schiff base potentially occurs, resulting in the formation of Amadori compounds (Wang, 2019; Jo, Park, Lee, & Lee, 2023). During the second stage, Amadori compounds undergo reactions involving retroaldolisation, cleavage, reduction, and hydrolysis when subjected to heat. These reactions lead to the formation of α-dicarbonyl compounds (α-DCs), such as glyoxal, methylglyoxal, and diacetyl (Supplementary Fig. 1) (Cha, Debnath, & Lee, 2019). Numerous factors potentially influence the generation of volatile compounds during the Maillard reaction. Notably, pH plays a crucial role in determining the specific pathway of the Maillard reaction, consequently influencing the aroma profile of the end-product. Under alkaline conditions, Amadori compounds degrade into α-DCs, whereas in acidic conditions, furfurals and their derivatives are formed (Lee, Yim, Lee, & Lee, 2023). Previous studies have attempted to minimise the flavour profile disparity between the Arabica and Robusta coffee varieties (Liu, Yang, Linforth, Fisk, & Yang, 2019). These studies focused on redirecting the acidity of green coffee beans to alter the Maillard reaction pathway, thereby influencing the production of volatile compounds during the roasting process (Lee, Yim, et al., 2023; Liu et al., 2019). In the ultimate stage, the highly reactive α-DCs give rise to nitrogen-containing compounds, such as pyrrole, imidazole, pyrazine, and melanoidins, via a process known as Strecker degradation (Wang, 2019). Intermediate substances of the Maillard reaction, such as α-DCs may lead to formation of neo-formed contaminants such as acrylamide, furan and 5-HMF. Roasting process conditions significantly impact the formation of neo-formed contaminants such as acrylamide and hydroxymethylfurfural (HMF), both of which are known to have potential carcinogenic properties (Barrios-Rodríguez, Gutiérrez-Guzmán, Amorocho-Cruz, Pedreschi, & Mariotti Celis, 2023).

α-DCs comprise two carbonyl compounds, and the most representative α-DCs are glyoxal (GO), methylglyoxal (MGO), and diacetyl (DA) (Marceau & Yaylayan, 2009). These α-DCs are yellow-coloured and highly volatile, reactive, and water-soluble (J. Cha et al., 2019; H. Lee, Yim, et al., 2023). Previous studies have demonstrated that α-DCs act as

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precursors to toxic compounds such as 4-methylimidazole, heterocyclic flavour compounds known for respiratory toxicity (Jang, Jiang, Hengel, & Shibamoto, 2013; Jeong, Cha, & Lee, 2017). Furthermore, α-DCs undergo reactions with free amino groups, resulting in the formation of advanced glycation end-products, which are linked to kidney and diabetic disorders (Choi, Noh, Lee, Lee, & Lee, 2023; Lee, Lee, et al., 2023). Protein and nucleotide glycation by GO and MGO damages the proteome, genome, and lipidome (Thornalley, 2008). GO inhalation irritates the eyes and respiratory organs (Jiang, Hengel, Pan, Seiber, & Shibamoto, 2013). MGO and DA have been implicated as potential contributors to both lactose intolerance and irritable bowel syndrome (Campbell et al., 2010).

Proteins are essential reactants in the Maillard reaction, which plays a pivotal role in developing the flavor, color, and aroma of roasted coffee. The studying the kinetics of protein degradation provides how different roasting methods influence the rate and extent of the Maillard reaction. This understanding allows to optimize roasting conditions to enhance desirable sensory attributes while minimizing the formation of harmful byproducts such as α-dicarbonyl compounds.

The traditional thermal process of coffee bean roasting involves using hot air to uniformly heat the surface of the beans, promoting efficient and uniform heat transfer to the core of the bean (Schenker, Handschin, Frey, Perren, & Escher, 2000). Previous studies have revealed that the mode of heat transfer during coffee bean roasting significantly influences the flavour profile of espresso coffee (Gancarz et al., 2022; Oliveros, Hernández, Sierra-Espinosa, Guardián-Tapia, & Pliego-Solórzano, 2017). In pan roasting, heat is directly applied from a flame or hot plate via surface contact. Stirring or mixing is employed to achieve uniform and rapid distribution of heat energy, expediting the roasting process (Sruthi, Premjit, Pandiselvam, Kothakota, & Ramesh, 2021). Air frying is a cooking method based on rapid air technology. During this process, a heating element at the top of the appliance emits heat downward, while a fan circulates air swiftly around the food, ensuring heating from all directions (Devi, Zhang, Ju, & Bhandari, 2021). The air frying, while it is more commonly used in a home-roasting context, it was included in this study due to its growing popularity and its potential benefits in reducing harmful compounds during roasting. Additionally, air frying presents a unique heat transfer mechanism that combines convection with rapid air circulation, making it an interesting method to study in comparison to more traditional roasting techniques. Although it may not have direct industrial relevance, air frying represents an emerging method that could influence future developments in small-scale roasting technologies.

Previous studies have reported the kinetic modelling of coffee Maillard reaction products, namely, furan, 5-hydroxymethylfurfural, and acrylamide (Albouchi & Murkovic, 2018; Zhu et al., 2022). They have also used kinetic modelling to analyse the concentration of proteins, as Maillard reaction reactants, and α-DCs in glucose–leucine cake as a first-order reaction (Helou, Jacolot, Niquet-Léridon, Gadonna-Widehem, & Tessier, 2016; Lee et al., 2022; Taş & Gökmen, 2017). However, no study has performed kinetic modelling of α-DCs in coffee prepared via various roasting methods.

This study aimed to analyse Maillard reaction products and proteins under different roasting conditions (three roasting methods, three roasting temperatures, and 10 roasting times). These compounds were selected since they are key markers of the Maillard reaction, affecting both the sensory profile and safety of roasted coffee. Studying their kinetics helps optimize roasting conditions across different methods. It provides the researchers with insight into the prediction and regulation of α-DC, melanoidin, and protein concentrations in roasted coffee beans based on various roasting methods.

2. Materials and methods

2.1. Chemical reagents and materials

Coffea arabica green coffee beans from Brazil were purchased from Trini Coffee Roastery Cafe in Ilsan, Gyeonggi-do, Korea. Before undergoing analysis, the green coffee beans were stored at room temperature in the dark.

For α-DC analysis, glyoxal (GO) (purity 39%) was acquired from Daejung Chemical & Metals Co., Ltd. (Gyeonggi-do, Korea). Methylglyoxal (MGO) (purity 40%), diacetyl (DA) (purity 97%), 1-methyl pyrazole, and o-phenylenediamine were secured from Sigma-Aldrich Chemical Co. (St. Louis, MO, USA). HPLC-grade water and ethyl acetate were purchased from J.T. Baker (Philipsburg, NJ, USA).

For crude protein analysis, sulfuric acid was acquired from Sigma-Aldrich Chemical Co. (St. Louis, MO, USA). Sodium hydroxide (NaOH) and hydrochloric acid (HCl) were secured from Daejung Chemical & Metals Co., Ltd. (Gyeonggi-do, Korea). Boric acid was purchased from Junsei Chemical Co., Ltd. (Tokyo, Japan).

2.2. Coffee sample preparation

A total of 90 combinations of coffee samples according to the roasting conditions were subjected to α-DC, melanoidin, and crude protein analyses. Briefly, 100-g samples of coffee beans were roasted using a hot air roaster (CBR-101A; Gene Café, Korea), a pan (MK-300; J. Roaster, China), or an air fryer (MEA-TC1000W; Magic Chef, China) at 240 °C, 220 °C, and 200 °C for 1, 3, 6, 12, 15, 18, 21, 24, 27, and 30 min. In the case of the MK-300 system, temperature control was adjusted using an infrared thermometer to maintain a consistent temperature throughout the roasting process. For the other two pieces of equipment, roasting began once a stable temperature was reached after preheating, ensuring that the temperature remained consistent from start to finish. The roasted coffee beans were ground with a grinder (Delonghi KG200; Treviso, Italy) for 30 s twice. Thereafter, 12.5-g samples of ground coffee bean powder and 100 mL of distilled water were used for extraction with an espresso coffee machine (Delonghi ECO311; Treviso, Italy).

2.3. Validation of α-DC samples

To validate α-DC concentration, the limit of detection (LOD), limit of quantification (LOQ), recovery, and precision (intraday and interday) were calculated based on Food and Drug Administration standards (FDA, 2019). To measure linearity, an eight-point (0, 0.1, 0.5, 1, 5, 10, and 50 μg/mL) calibration curve was constructed and used to calculate the coefficient of determination (R^2). As an internal standard, 1-methyl pyrazole was employed, and 500 ppm 1-methyl pyrazole solution was added to each vial. Three-Sigma method equations were used to calculate the LOD: standard deviation/slope × 3.3. The LOQ was calculated using the 10-Sigma method: standard deviation/slope × 10). The standard deviations of the LOD and LOQ calculations were obtained by measuring the blank value five times. Recovery (%) was measured by adding 5, 10, and 50 μg/mL of α-DC. To measure precision, intraday was measured by repeating the analysis five times per day, and interday was obtained by conducting analyses over 5 days. The relative standard deviation was measured five times at three concentrations (5, 10, and 50 μg/mL).

2.4. Analysis of α-DCs in coffee prepared using hot air, pan roasting, and air frying via gas chromatography–nitrogen-phosphorous detection (GC–NPD)

In a previous study, liquid–liquid extraction was used to derivatise the α-DCs in coffee to quinoxalines using o-phenylenediamine (Lee, Yim, et al., 2023). A 20-mL headspace vial was filled with a mixture comprising 3 mL of espresso coffee and 2 mL of o-phenylenediamine

solution (0.01 g/mL). After adding 0.1 mol/L NaOH solution to adjust the combined solution's pH to 12, it was stirred for 2 h at 405×*g*. After derivatisation, 5 mL of ethyl acetate and 20 μL of 1-methyl pyrazole (internal standard, 500 μg/mL) were added to the combined solution. Thereafter, the derivatised α-DCs were extracted to ethyl acetate by shaking for 5 min. After shaking, the solutions were centrifuged (1580R-refrigerated large-capacity centrifuge; Gyrozen Co., Ltd., Gimpo, Korea) at 2160×*g* and 4 °C for 10 min. After removing the bottom phase, the supernatant was quantified and analysed using GC–NPD on 1 μL of supernatant.

An Agilent Technologies (Palo Alto, CA, USA) GC–NPD (Agilent 6890N) system equipped with a DB-WAX column (60 m × 0.30 mm × 0.25 μm, film thickness; J&W Scientific, Folsom, CA) was used for α-DC analysis. As the carrier gas, helium was used at a constant flow rate of 1.5 mL/min in splitless mode. The injector was applied in splitless injection mode at 260 °C. The oven program was initiated at 40 °C for 2 min, followed by a gradual increase to 170 °C; maintained for 15 min; and completed with a post-run at 230 °C for 10 min. The detector temperature was set at 300 °C. Hydrogen and pure air, at flow rates of 120 and 2 mL/min, respectively, were employed for ionisation. A Blos Bead detector (Agilent Technologies, Palo Alto, CA, USA) was used for NPD. A 20–23-pA range was used as the signal base.

2.5. Analysis of melanoidins in coffee

A high-molecular-weight molecule known as melanoidin, along with groups of chromophores with a specific absorbance at 420 nm, are produced during the Maillard reaction. In previous studies, melanoidin was analysed using a UV spectrometer (Feng et al., 2023). The following method was employed to measure absorbance at 420 nm. In a 50-mL conical tube, 1.5 mL of espresso coffee and 30 mL of distilled water were mixed. An ultrasonic bath (Branson Ultrasonic Cleaner; Danburry, CT, USA) was used to sonicate the conical tube for 10 min. Subsequently, samples were extracted into 50-mL conical tubes by centrifuging at 4 °C for 10 min at 14,000×*g*. The absorbance of the melanoidins was measured at 420 nm using a UV/Vis spectrophotometer (UV–1800; Shimadzu Corporation, Kyoto, Japan). Analyses were conducted in triplicate.

Considering the complexity of brown-coloured compounds, quantifying the composition of melanoidins from the Maillard reaction is difficult (Echavarría, Pagán, & Ibarz, 2012). In a previous study, melanoidin concentration was calculated using the Lambert–Beer equation, which has an extinction coefficient of 282 L/mol same as current study (Knol, Linssen, & van Boekel, 2010; Zhang et al., 2019).

2.6. Analysis of crude proteins in coffee

Digestion and protein–nitrogen distillation apparatus (DNP-1500-MP) were used for the analysis. Samples of approximately 0.5 g were weighed. Several studies using the Kjeldahl method to analyse protein content have been reported several times (Bettaieb et al., 2024; Prandi et al., 2021; Zhu et al., 2021). The weighed samples were subsequently transferred to a Kjeldahl digestion flask containing a catalyst (1 g Cu_2SO_4 and 4 g K_2SO_4). Before digestion, 12 mL of sulfuric acid was added to the Kjeldahl digestion flask. After 2 h of digestion and cooling to room temperature, boric acid solution (25 mL) was added to a flask containing 30 mL of the digested sample for distillation, and 50 mL of NaOH was distilled into the flask using protein–nitrogen distillation apparatus. Crude protein content was determined via titration with standardised HCl until a mixed indicator endpoint was reached.

2.7. Kinetic modelling

The following equation model was applied to determine the Maillard reaction product and crude protein concentrations in coffee beans processed via various roasting methods, at different temperatures, and over varying time periods:

$$\frac{d[C]}{dt} = k[C]^n \tag{1}$$

When integrated for first-order reactions (Equation (1)), the resulting equations represent the respective first-order mass formation models.

$$[C] = [C]_0\, e^{kt} \tag{2}$$

Based on Equation (2), the kinetic modelling of α-DCs, melanoidins, and proteins in coffee beans during roasting can be explained using Equations (3) and (4):

$$\text{Formation stage}: C_t = C_1 e^{k_1 t} \tag{3}$$

$$\text{Degradation stage}: C_t = C_2 e^{-k_2 t} \tag{4}$$

Where C_t represents the concentration at roasting time (t), k is the reaction rate constant (1/min), and C_1 and C_2 represent the initial formation- and degradation-stage concentrations, respectively. First-order kinetic modelling was performed using Origin 2023 (MicroCal Software, Inc., Northampton, MA, USA).

2.8. Arrhenius plot

The Arrhenius equation (Equation (5)) was used to ascertain how temperature affects the different roasting methods.

$$\ln(\mathrm{k}) = \ln(\mathrm{A}) - \frac{Ea}{R}\left(\frac{1}{T}\right) \tag{5}$$

Where k is the reaction rate constant (mol/L•h), E_a is the apparent activation energy (kJ/mol), T is the absolute temperature (K), R is the gas constant (8.314 J/mol), and A is the frequency factor. The apparent reaction E_a values for α-DCs, melanoidins, and crude proteins were calculated using simple linear regression of ln(k) against 1/T based on the least-squares method.

2.9. Measurement of colour value

Colour changes in the ground roasted coffee bean samples were measured using a colour meter (Nipon Denshoku Industries, Co., Ltd., Tokyo, Japan) in reflection mode, with the samples placed in petri dishes. The L* [light (L* = 100) and dark (L* = 0)], a* [redness (+) and greenness (−)], and b* [yellowness (+) and blueness (−)] values were measured. The colour differences among the ground roasted coffee bean samples were indicated using a colour difference indicator.

2.10. Statistical analysis

Regarding to the statistical analysis, a statistical expert was consulted to confirm the normalcy of the data. All results are expressed as the mean ± standard deviation (SD) based on triplicate analyses. Using IBM SPSS Statistics (version 23; IBM, Chicago, USA), this study conducted one-way analysis of variance and Duncan's multiple range test to detect significant differences ($p < 0.05$) among triplicate samples and identify the influence of other variables.

3. Results and discussion

3.1. Validation of α-DC analysis

Validation of α-DC analysis was shown in Supplementary Table 1. Regarding linearity, the R^2 values of the α-DCs GO, MGO, and DA were 0.9990, 0.9992, and 0.9995, respectively. The LOD values of GO, MGO, and DA were 0.05, 0.04, and 0.04 μg/mL, respectively. The LOQ values of GO, MGO, and DA were 0.18, 0.12, and 0.13 μg/mL, respectively. The

recovery rate ranged from 95.7% (DA) to 120.32% (MGO), while the intraday and interday precision results (RSD) ranged from 1.42% (DA) to 3.68% (MGO) and 0.78% (MGO) to 9.77% (DA), respectively. The RSD value was <20% and recovery rate range 80–120%, thus satisfying Food and Drug Administration standards (FDA, 2019). A previous study analysing α-DCs in coffee reported a recovery range of 88.0–119.4% and RSD range of 0.46–12.08% (Hyong, Chu, Park, Park, & Lee, 2021). This study's validation results are consistent with those of a previous study and remained within acceptable range according to the Food and Drug Administration guidelines.

3.2. Analysis of α-DCs in coffee

In total, 90 samples were used to analyse α-DC concentration in coffee beans processed via various roasting methods, under different conditions. Samples were analysed using GC–NPD to verify the effects of the roasting method, temperature, and time. Table 1 shows the α-DC concentrations (GO, MGO, and DA) in the 90 coffee samples. Interestingly, at the same roasting temperature, α-DC concentration increased with roasting time, reached a maximum, and subsequently decreased. This is because as roasting proceeds, the more the Maillard reaction progresses, resulting in more α-DCs being converted to volatile compounds and pigments such as melanoidins (Kwon et al., 2021).

In the hot air-roasted coffee samples, the total α-DC concentration ranged from 1.29 to 71.24 μg/mL. At roasting temperatures of 240, 220, and 200 °C, the roasting times that elicited the highest α-DC concentrations were 12, 15, and 21 min, respectively. These results indicate that the higher the roasting temperature, the faster the α-DCs are produced. The highest α-DC concentrations obtained at roasting temperatures of 240, 220, and 200 °C were 64.54, 67.17, and 71.24 μg/mL, respectively. A previous study demonstrated that the concentration of α-DCs, including GO, MGO, and DA, was significantly reduced as the roasting temperature increased (Kwon et al., 2021). This is because the Maillard reaction produces α-DCs in its intermediate stage, and higher roasting temperatures cause the Maillard reaction to become more active, converting α-DCs into other substances, such as melanoidins and volatile compounds. (Bi et al., 2024). Pan and air fryer roasting also exhibit this tendency.

In the pan-roasted coffee samples, the total α-DC concentration ranged from 1.76 to 68.58 μg/mL. At roasting temperatures of 240, 220, and 200 °C, the roasting times that yielded the highest α-DC concentrations were 6, 12, and 15 min, respectively. The highest α-DC concentrations obtained at roasting temperatures of 240, 220, and 200 °C were 59.31, 62.02, and 68.58 μg/mL, respectively. These results demonstrate that pan roasting accelerates the roasting process more rapidly than hot air and air fryer roasting. In hot air and air fryer roasting, convection heat transfer is based on hot air rising from a hot surface toward the coffee beans, whereas in pan roasting, conduction heat transfer is based on the coffee beans coming into contact with a warm surface (Macheiner, Schmidt, Karpf, & Mayer, 2021). Unlike hot air and air fryer roasting, owing to differences in the main heat transfer mechanism, pan roasting proceeds rapidly as the coffee beans are in direct contact with the surface of the hot plate (Sruthi et al., 2021).

In the air fryer-roasted coffee samples, the total α-DC concentration ranged from 1.24 to 77.00 μg/mL. The highest α-DC concentrations obtained at roasting temperatures of 240, 220, and 200 °C were 69.77, 71.69, and 77.00 μg/mL, respectively. The rapid air process employed in the air fryer uses a heating element at the top of the appliance to radiate heat downward, while a fan rapidly circulates air around the food inside to heat it uniformly from all sides (Devi et al., 2021). The common characteristic between the roaster and air fryer is that hot air is circulated to evenly heat the coffee beans from all angles; nevertheless, no study has applied air fryer roasting to coffee. Previous studies have used an air fryer to roast potatoes and chicken nuggets, replacing traditional roasting methods (Cao et al., 2020; Shaker, 2015). Air frying has been found to be a more suitable method for producing healthier fried potatoes as it has a lower peroxide value than traditional frying methods (Shaker, 2015).

As shown in Table 1, at roasting temperatures of 240, 220, and 200 °C, the roasting times that elicited the highest α-DC concentrations were 12, 15, and 18 min, respectively. This tendency is similar to that exhibited by hot air roasting. These results suggest that the rate of the Maillard reaction is similar between hot air and air fryer roasting. According to Table 1, air fryer roasting yields a lower α-DC concentration than hot air roasting after 21 min at 200 °C. Therefore, these results suggest that air fryer roasting can be applied as a roasting method that produces fewer toxic compounds than hot air roasting, which is a traditional roasting method.

3.3. Analysis of melanoidins in coffee

During the Maillard reaction, various flavour, toxic, and colour compounds are formed. Among them is melanoidin, which is a brown-coloured compound that is formed in the final stage of the Maillard reaction (Lee, Yim, et al., 2023). The concentration of melanoidins in the 90 coffee samples is shown in Table 1. In the 90 coffee samples, melanoidin concentration ranged from 0.013 to 0.422 mmol/L. At roasting temperatures of 240, 220, and 200 °C, the higher the temperature, the higher the melanoidin concentration. Moreover, in hot air, pan, and air fryer roasting, the longer the roasting time, the higher the concentration, regardless of temperature. In a previous study, the absorbance at 420 nm was higher when roasting was performed on green coffee beans to achieve dark roasting (Alongi et al., 2021). These results reveal that a higher temperature and longer roasting time cause the Maillard reaction to occur more rapidly, resulting in the increased formation of melanoidins.

In the hot air-roasted coffee samples, melanoidin concentration ranged from 0.014 to 0.305 mmol/L. At roasting temperatures of 240, 220, and 200 °C, the roasting time that resulted in the highest melanoidin concentration was 30 min. These results indicate that the longer the roasting time, the more the Maillard reaction progresses, thus increasing the production of melanoidins, products of the Maillard reaction's final stage. In addition, the pan- and air fryer-roasted samples tended to exhibit similar results. Compared with 1-min roasting at temperatures of 240, 220, and 200 °C, 30-min roasting using hot air increased the melanoidin content of coffee by 1170.8, 1090.0, and 992.9%, respectively.

In the pan-roasted coffee samples, melanoidin concentration ranged from 0.014 to 0.422 mmol/L. Compared with 1-min roasting at temperatures of 240, 220, and 200 °C, 30-min roasting using a pan increased the melanoidin content of coffee by 1355.2, 1505.3, and 1328.6%, respectively. These results demonstrate that pan roasting is faster than hot air roasting.

In the air fryer-roasted coffee samples, melanoidin concentration ranged from 0.013 to 0.298 mmol/L. After 30 min of air fryer roasting at 240, 220, and 200 °C, the melanoidin concentration in the coffee samples increased by 1092.0, 1081.3, and 1084.6%, respectively, compared with that after roasting for 1 min. On comparing the three roasting methods, the rate of melanoidin formation was in the following order: pan roasting > hot air roasting > air fryer roasting. Pan roasting predictably causes the highest Maillard reaction rate, exhibiting consistency with the α-DC results. With the same degree of roasting, the brown pigments typically have a positive visual impact on customers. When attempting to increase the melanoidin concentration, pan roasting is reportedly the most suitable roasting method (Mesías & Delgado-Andrade, 2017).

3.4. Analysis of crude proteins in coffee

The quantitative crude protein concentrations are shown in Table 1. The crude protein concentration after 1 min of roasting ranged from 14.92 to 16.05 g/100 g, displaying similar results to those yielded with

Table 1
Concentration of α-DCs, melanoidins, and crude protein in coffee samples.

Roasting method	Roasting time (min)	Roasting temperature								
		200 °C			220 °C			240 °C		
		α-DCs (μg/mL)	Melanoidins (mmol/L)	Protein Concentration (g/100g)	α-DCs (μg/mL)	Melanoidins (mmol/L)	Protein Concentration (g/100g)	α-DCs (μg/mL)	Melanoidins (mmol/L)	Protein Concentration (g/100g)
Hot air	1	1.29 ± 0.06^{a}	0.014 ± 0.004^{a}	15.54 ± 0.72^{k}	1.33 ± 0.07^{a}	0.020 ± 0.004^{a}	15.54 ± 0.72^{k}	3.45 ± 0.32^{a}	0.024 ± 0.005^{a}	15.47 ± 0.27^{m}
	3	3.93 ± 0.03^{b}	0.023 ± 0.004^{ab}	15.04 ± 0.45^{ijk}	4.58 ± 0.32^{b}	0.027 ± 0.005^{ab}	15.04 ± 0.45^{ijk}	6.41 ± 0.30^{b}	0.035 ± 0.005^{b}	14.79 ± 0.48^{lm}
	6	10.09 ± 0.13^{d}	0.043 ± 0.004^{bc}	14.86 ± 0.47^{ijk}	14.64 ± 0.25^{d}	0.036 ± 0.001^{b}	14.86 ± 0.47^{ijk}	27.63 ± 0.29^{f}	0.059 ± 0.002^{c}	13.79 ± 0.28^{k}
	12	21.49 ± 1.03^{f}	0.083 ± 0.001^{cd}	14.06 ± 0.26^{hij}	28.41 ± 1.48^{f}	0.074 ± 0.001^{cd}	14.06 ± 0.26^{hij}	64.54 ± 3.50^{q}	0.100 ± 0.004^{d}	13.03 ± 0.47^{ij}
	15	32.46 ± 1.16^{hi}	0.085 ± 0.001^{cd}	13.59 ± 0.27^{ghi}	67.17 ± 1.21^{p}	0.097 ± 0.002^{d}	13.59 ± 0.27^{ghi}	55.45 ± 2.41^{o}	0.156 ± 0.005^{e}	12.98 ± 0.47^{ij}
	18	56.54 ± 1.06^{o}	0.090 ± 0.001^{d}	13.24 ± 0.47^{fgh}	58.38 ± 1.92^{n}	0.094 ± 0.002^{d}	13.24 ± 0.47^{fgh}	38.91 ± 1.31^{l}	0.168 ± 0.003^{ef}	12.26 ± 0.28^{fg}
	21	71.24 ± 3.27^{s}	0.098 ± 0.002^{d}	12.76 ± 0.27^{def}	51.04 ± 3.74^{kl}	0.109 ± 0.001^{de}	12.76 ± 0.27^{def}	36.08 ± 3.52^{jk}	0.191 ± 0.006^{g}	12.62 ± 0.27^{gh}
	24	67.79 ± 4.95^{q}	0.117 ± 0.002^{de}	12.43 ± 0.27^{def}	48.09 ± 1.17^{j}	0.115 ± 0.003^{de}	12.43 ± 0.27^{def}	32.43 ± 3.50^{hi}	0.192 ± 0.001^{g}	11.40 ± 0.27^{cde}
	27	63.00 ± 1.41^{p}	0.133 ± 0.002^{e}	12.19 ± 0.47^{cde}	42.57 ± 1.93^{i}	0.153 ± 0.005^{efg}	12.19 ± 0.47^{cde}	31.25 ± 1.47^{h}	0.245 ± 0.002^{h}	10.90 ± 0.27^{bc}
	30	54.70 ± 1.82^{mo}	0.153 ± 0.003^{f}	11.32 ± 0.27^{bc}	34.73 ± 1.14^{gh}	0.238 ± 0.005^{h}	11.32 ± 0.27^{bc}	30.56 ± 1.71^{ef}	0.305 ± 0.003^{i}	10.87 ± 0.54^{bc}
Pan	1	1.86 ± 0.08^{a}	0.014 ± 0.001^{a}	14.89 ± 0.27 ^{ijk}	1.76 ± 0.07^{a}	0.019 ± 0.001^{a}	14.89 ± 0.27^{ijk}	6.09 ± 0.21^{b}	0.029 ± 0.001^{a}	14.92 ± 0.27^{lm}
	3	6.51 ± 0.20^{c}	0.027 ± 0.001^{ab}	13.79 ± 0.27^{ghi}	8.09 ± 0.32^{c}	0.032 ± 0.001^{b}	13.79 ± 0.27^{ghi}	30.81 ± 1.41^{h}	0.043 ± 0.011^{bc}	13.76 ± 0.25^{k}
	6	25.64 ± 0.91^{g}	0.044 ± 0.001^{bc}	12.92 ± 0.27^{def}	28.84 ± 1.25^{f}	0.053 ± 0.004^{c}	12.92 ± 0.27^{def}	59.31 ± 3.33^{p}	0.069 ± 0.011^{cd}	12.87 ± 0.27^{hi}
	12	50.88 ± 2.50^{l}	0.077 ± 0.001^{h}	12.19 ± 0.47^{cde}	62.02 ± 3.48^{o}	0.082 ± 0.002^{d}	12.19 ± 0.47^{cde}	42.18 ± 1.31^{m}	0.119 ± 0.001^{d}	12.11 ± 0.28^{efg}
	15	68.58 ± 1.29^{qr}	0.083 ± 0.001^{cd}	12.09 ± 0.47^{cde}	55.32 ± 1.21^{m}	0.112 ± 0.001^{de}	12.09 ± 0.47^{cde}	38.92 ± 1.82^{l}	0.175 ± 0.025^{f}	11.96 ± 0.27^{def}
	18	53.10 ± 1.49^{lm}	0.087 ± 0.003^{cd}	11.90 ± 0.73^{cd}	52.90 ± 1.92^{lm}	0.123 ± 0.001^{e}	11.90 ± 0.73^{cd}	31.42 ± 1.55^{h}	0.202 ± 0.002^{g}	10.98 ± 0.27^{bc}
	21	48.73 ± 2.37^{kl}	0.095 ± 0.011^{d}	12.09 ± 0.25^{b}	48.29 ± 3.74^{j}	0.142 ± 0.001^{ef}	11.33 ± 0.73^{bc}	28.07 ± 1.82^{fg}	0.231 ± 0.005^{h}	10.31 ± 0.28^{ab}
	24	41.76 ± 2.84^{k}	0.132 ± 0.002^{e}	11.96 ± 0.25^{ab}	40.53 ± 1.17^{i}	0.162 ± 0.001^{fg}	10.83 ± 0.55^{ab}	25.00 ± 1.53^{de}	0.246 ± 0.001^{h}	10.36 ± 0.28^{ab}
	27	39.54 ± 1.50^{jk}	0.138 ± 0.002^{e}	11.50 ± 0.27^{ab}	33.61 ± 1.93^{g}	0.187 ± 0.001^{g}	10.49 ± 0.48^{ab}	24.34 ± 1.48^{de}	0.303 ± 0.001^{i}	10.25 ± 0.23^{ab}
	30	36.23 ± 2.52^{ij}	0.200 ± 0.003^{g}	11.15 ± 0.49^{a}	27.10 ± 1.14^{f}	0.305 ± 0.001^{i}	10.12 ± 0.28^{a}	15.50 ± 0.99^{c}	0.422 ± 0.002^{j}	9.88 ± 0.47^{a}
Air fryer	1	1.70 ± 0.04^{a}	0.013 ± 0.001^{a}	16.05 ± 0.27^{i}	1.24 ± 0.08^{a}	0.016 ± 0.002^{a}	15.70 ± 0.27^{k}	3.23 ± 0.08^{a}	0.025 ± 0.001^{a}	15.44 ± 0.55^{m}
	3	3.30 ± 0.44^{b}	0.024 ± 0.002^{ab}	15.73 ± 0.48^{gh}	5.10 ± 0.60^{b}	0.028 ± 0.003^{ab}	15.28 ± 0.25^{jk}	5.99 ± 0.22^{b}	0.038 ± 0.011^{b}	14.59 ± 0.48^{l}
	6	17.58 ± 1.26^{e}	0.039 ± 0.002^{b}	15.04 ± 0.54^{fg}	23.00 ± 1.59^{e}	0.035 ± 0.005^{b}	15.07 ± 0.27^{ijk}	23.75 ± 1.57^{d}	0.054 ± 0.010^{c}	13.57 ± 0.27^{jk}
	12	29.87 ± 2.77^{h}	0.088 ± 0.004^{cd}	14.44 ± 0.71^{ef}	47.93 ± 2.78^{j}	0.068 ± 0.003^{cd}	14.25 ± 0.28^{ij}	69.77 ± 3.79^{r}	0.146 ± 0.001^{e}	12.93 ± 0.27^{hi}
	15	57.00 ± 1.66^{o}	0.083 ± 0.003^{cd}	14.18 ± 0.27^{e}	71.69 ± 3.78^{q}	0.099 ± 0.005^{d}	13.51 ± 0.27^{ghi}	59.57 ± 1.51^{p}	0.161 ± 0.023^{ef}	12.81 ± 0.48^{hi}
	18	77.00 ± 1.22^{t}	0.087 ± 0.001^{cd}	13.87 ± 0.27^{de}	55.18 ± 4.21^{m}	0.089 ± 0.002^{d}	13.17 ± 0.27^{fgh}	48.83 ± 1.32^{n}	0.175 ± 0.002^{ef}	12.28 ± 0.48^{fg}
	21	64.59 ± 4.64^{pq}	0.095 ± 0.005^{d}	13.53 ± 0.44^{de}	49.71 ± 1.82^{jk}	0.105 ± 0.001^{de}	12.85 ± 0.28^{efg}	40.29 ± 1.08^{lm}	0.202 ± 0.005^{g}	12.02 ± 0.27^{efg}
	24	57.54 ± 1.89^{o}	0.093 ± 0.002^{d}	13.33 ± 0.38^{d}	41.36 ± 1.37^{i}	0.117 ± 0.002^{de}	12.45 ± 0.28^{def}	36.59 ± 1.23^{k}	0.195 ± 0.003^{g}	11.30 ± 0.27^{cde}

(*continued on next page*)

Table 1 (*continued*)

Roasting method	Roasting time (min)	Roasting temperature								
		200 °C			220 °C			240 °C		
		α-DCs (μg/mL)	Melanoidins (mmol/L)	Protein Concentration (g/100g)	α-DCs (μg/mL)	Melanoidins (mmol/L)	Protein Concentration (g/100g)	α-DCs (μg/mL)	Melanoidins (mmol/L)	Protein Concentration (g/100g)
	27	51.73 ± 3.25^{l}	0.132 ± 0.002^{e}	12.93 ± 0.28^{cd}	40.14 ± 2.03^{i}	0.147 ± 0.004^{efg}	12.43 ± 0.23^{def}	34.10 ± 1.10^{ij}	0.250 ± 0.003^{h}	11.21 ± 0.27^{cd}
	30	47.45 ± 2.84^{kl}	0.154 ± 0.003^{f}	12.59 ± 0.49^{cd}	36.61 ± 1.20^{h}	0.189 ± 0.002^{g}	12.19 ± 0.47^{cde}	30.11 ± 0.88^{gh}	0.298 ± 0.002^{i}	11.02 ± 0.47^{bc}

1) All values are shown as mean ± S.D. (standard deviation) (n = 3).
2) Lowercase letters (series "a-t") indicate significant ($p < 0.05$) differences in the same column.

green coffee beans in another study (Alamri, Rozan, & Bayomy, 2022). In the hot air-roasted coffee samples, the crude protein concentration ranged from 10.87 to 15.82 g/100 g. At roasting temperatures of 240, 220, and 200 °C, the roasting time that elicited the lowest crude protein concentration was 30 min. These results indicate that the longer the roasting time, the more the crude proteins are degraded. Furthermore, in the pan- and air fryer-roasted samples, the results tended to be the same. Compared with 1-min roasting at 240, 220, and 200 °C, 30-min roasting using hot air decreased the crude protein content of coffee by 29.7, 27.1, and 22.0%, respectively.

In the pan-roasted coffee samples, the crude protein concentration ranged from 9.88 to 15.01 g/100 g. Compared with 1-min roasting at 240, 220, and 200 °C, 30-min roasting using a pan decreased the crude protein content of coffee by 33.8, 32.1, and 25.7%, respectively. According to these results, pan roasting elicits quicker results than hot air roasting. In the air fryer-roasted coffee samples, the crude protein concentration ranged from 10.87 to 15.82 g/100 g. The crude protein concentration of the coffee samples that were roasted using an air fryer for 30 min at 240, 220, and 200 °C decreased by 28.6, 33.8, and 21.6%, respectively, compared with that of the coffee samples roasted for 1 min.

On comparing the three roasting methods, the degree of crude protein decomposition predictably occurs in the following order: pan roasting > hot air roasting > air fryer roasting. Moreover, in terms of crude protein degradation, pan roasting elicited the highest Maillard reaction rate, exhibiting consistency with the results for α-DCs and melanoidins. Based on the α-DC, melanoidin, and crude protein results, pan roasting was confirmed to cause the highest Maillard reaction rate. A high Maillard reaction rate implies that Strecker degradation occurs relatively fast, with more volatile compounds being formed. Evidently, among the three roasting methods, pan roasting can improve volatile compound content (Das, Duarah, & Purkait, 2023). As sugars caramelize, they convert into furfural and furan, reducing the sugars available for the Maillard reaction. This limits the interaction between sugars and proteins during roasting. Additionally, since α-DCs are formed from both the Maillard reaction and sugar degradation, more α-DCs were detected than would result solely from the Maillard reaction. Since α-DCs are formed from both the Maillard reaction and sugar degradation, more α-DCs were detected than would result solely from the Maillard reaction. The kinetics of α-dicarbonyl compound formation and its relationship with the production of acrylamide and other harmful neo-formed contaminants have been well-studied in recent research, suggesting important mitigation strategies (Barrios-Rodríguez et al., 2022, 2023). Therefore, it is crucial to further investigate how the kinetics of carbonyl and melanoidin formation or elimination in roasted coffee are linked to the kinetics of formation or degradation of neo-formed contaminants like acrylamide and furan. Both acrylamide and furan are known to form through Maillard intermediates such as α-dicarbonyls and melanoidins, but their exact kinetics under varying roasting conditions require more detailed exploration to develop effective mitigation strategies.

3.5. Kinetic modelling

The kinetic model of two first-order processes (Equations (3) and (4)) is presented in Table 2 to explain the formation and degradation rate constants of α-DCs. The changes in the α-DC concentration of roasted coffee can be attributed to various roasting methods, temperatures, and times, as shown in Table 1. In the hot air-roasted samples, the rate constants for the formation stage ranged from 0.185 to 0.269 min^{-1}, while those for the degradation stage ranged from 0.0290 to 0.0430 min^{-1}. In the pan-roasted samples, the rate constants ranged from 0.237 to 0.437 min^{-1} and 0.040 to 0.050 min^{-1} for the formation and degradation stages, respectively. In the air fryer-roasted samples, the rate constants for the formation stage ranged from 0.216 to 0.282 min^{-1}, while those for the degradation stage ranged from 0.040 to 0.0470 min^{-1}.

Table 2
First-order kinetic parameters fitted for α-DCs, melanoidins, and crude protein in coffee beans.

Roasting method	Compounds	Parameters	200 °C	220 °C	240 °C
Hot air	α-DCs	C_1	1.970	1.768	3.218
		k_1	0.185	0.249	0.269
		R^2	0.954	0.980	0.937
		C_2	133.140	123.820	96.981
		k_2	0.0290	0.0410	0.0430
		R^2	0.943	0.984	0.892
	Melanoidins	C_1	0.0489	0.0520	0.0712
		k_1	0.0726	0.0756	0.0815
		R^2	0.928	0.938	0.928
	Proteins	C_2	15.768	15.728	15.280
		k_2	0.0080	0.0100	0.0120
		R^2	0.986	0.987	0.955
Pan	α-DCs	C_1	2.790	2.419	5.210
		k_1	0.237	0.301	0.437
		R^2	0.900	0.901	0.903
		C_2	116.050	112.480	79.789
		k_2	0.0400	0.0450	0.0500
		R^2	0.938	0.948	0.981
	Melanoidins	C_1	0.0491	0.0579	0.0823
		k_1	0.0755	0.0819	0.0850
		R^2	0.952	0.952	0.958
	Proteins	C_2	14.853	14.410	14.388
		k_2	0.0100	0.0120	0.0140
		R^2	0.979	0.957	0.951
Air fryer	α-DCs	C_1	2.109	1.950	2.846
		k_1	0.216	0.264	0.282
		R^2	0.942	0.921	0.971
		C_2	152.360	125.640	116.820
		k_2	0.0400	0.0430	0.0470
		R^2	0.980	0.935	0.978
	Melanoidins	C_1	0.0469	0.0495	0.0729
		k_1	0.0730	0.0750	0.0795
		R^2	0.867	0.947	0.882
	Proteins	C_2	16.012	15.749	15.028
		k_2	0.008	0.009	0.011
		R^2	0.991	0.983	0.961

In all three roasting methods, the higher the temperature, the higher the rate constants for the formation and degradation stages. These results indicate that the higher the temperature, the faster the Maillard reaction progresses. On comparing the three roasting methods, pan roasting yielded the highest rate constant for both the formation and degradation stages, followed by air fryer and hot air roasting. Because conduction, which involves direct contact with the heating surface, is the main heat transfer mechanism in pan roasting, unlike in hot air and air fryer roasting, the Maillard reaction occurs more rapidly (Sruthi et al., 2021). In the pan-roasted samples, the rate constant values at 240, 220, and 200 °C for the formation stage were 62.6%, 21.0%, and 28.3% higher than those in the hot air-roasted samples, respectively. Moreover, the rate constant values at 240, 220, and 200 °C for the degradation stage were 16.3%, 9.8%, and 37.9% higher than those in hot air roasting, respectively. In the air fryer-roasted samples, the rate constant values at 240, 220, and 200 °C for the formation stage were 4.8%, 6.4%, and 17.2% higher than those in the hot air-roasted samples, respectively. Furthermore, the rate constant values at 240, 220, and 200 °C for the degradation stage were 9.3%, 4.9%, and 37.9% higher than those in the hot air-roasted samples, respectively. In all three roasting methods, the rate constant for the formation stage exceeded that for the degradation stage. These results demonstrate that α-DCs form more rapidly than they degrade.

Table 2 shows the kinetic model of first-order processes (Equation (3)) for the formation of melanoidins. The rate constants for the formation stage in the hot air-roasted samples ranged from 0.0726 to 0.0815 min^{-1}. Compared with those yielded in air fryer roasting, the rate constant values for the formation stage at 240, 220, and 200 °C were 0.7%, 1.3%, and 3.0% higher, respectively. In the pan-roasted samples, the rate constants for the formation stage ranged from 0.076 to 0.085 min^{-1}. The rate constant values at 240, 220, and 200 °C for the formation stage were 4.8%, 9.3%, and 6.8% higher than those yielded in air fryer roasting. In the air fryer-roasted samples, the rate constants for the formation stage ranged from 0.0730 to 0.0795 min^{-1}. In all three roasting methods, higher temperatures resulted in higher formation-stage rate constants.

The kinetic model of first-order processes (Equation (4)) reveals how crude proteins degrade. In the hot air-roasted samples, the rate constants for the degradation stage ranged from 0.008 to 0.011 min^{-1}. Compared with those yielded in air fryer roasting, the rate constant values for the degradation stage at 240 and 220 °C were 9.1% and 11.1% higher, respectively. The rate constants for the degradation stage in the pan-roasted samples ranged from 0.0100 to 0.0140 min^{-1}. The rate constant values at 240, 220, and 200 °C for the degradation stage were 25.0%, 33.3%, and 27.3% higher than those yielded in air fryer roasting. In the air fryer-roasted samples, the rate constants for the degradation stage ranged from 0.008 to 0.0120 min^{-1}.

On comparing the rate constants for the formation and degradation of α-DCs, formation of melanoidins, and degradation of crude proteins, pan roasting exhibited the highest rate constant value. This result is consistent with α-DC, melanoidin, and crude protein concentration predictions. Based on the rate constant results, pan roasting is the fastest roasting method among the three roasting methods because unlike hot air and air fryer roasting, the heat transfer method is conduction rather than convection (Sruthi et al., 2021).

3.6. Arrhenius plot

The α-DC concentration results from the three roasting methods were used to construct an Arrhenius plot. Fig. 1 displays the Arrhenius plots of ln k versus 1/T for the formation and degradation reactions of the Maillard reaction products and proteins in coffee beans during roasting. To obtain the E_a, a previous study used the slope derived by performing a linear regression analysis of each temperature (Kim, Lee, & Lee, 2022).

The E_a values for the α-DC formation stages of the hot air, pan, and air fryer roasting methods were 18.24, 30.71, and 13.38 kJ/mol, respectively. Moreover, the E_a values for the α-DC degradation stages of the hot air, pan, and air fryer roasting methods were 19.84, 10.65, and 8.37 kJ/mol, respectively. In hot air roasting, temperature was found to influence α-DC degradation more than α-DC formation. In contrast, pan and hot air roasting were observed to exert a greater temperature effect on α-DC formation than on α-DC degradation. Overall, air fryer roasting was found to exert the smallest temperature effect on α-DC formation and degradation. In hot air roasting, the E_a values for the formation and degradation stages were 36.3% and 137.1% higher than those in air fryer roasting, respectively. Furthermore, in pan roasting, the E_a values for the formation and degradation stages were 129.5% and 27.3% higher than those in air fryer roasting, respectively.

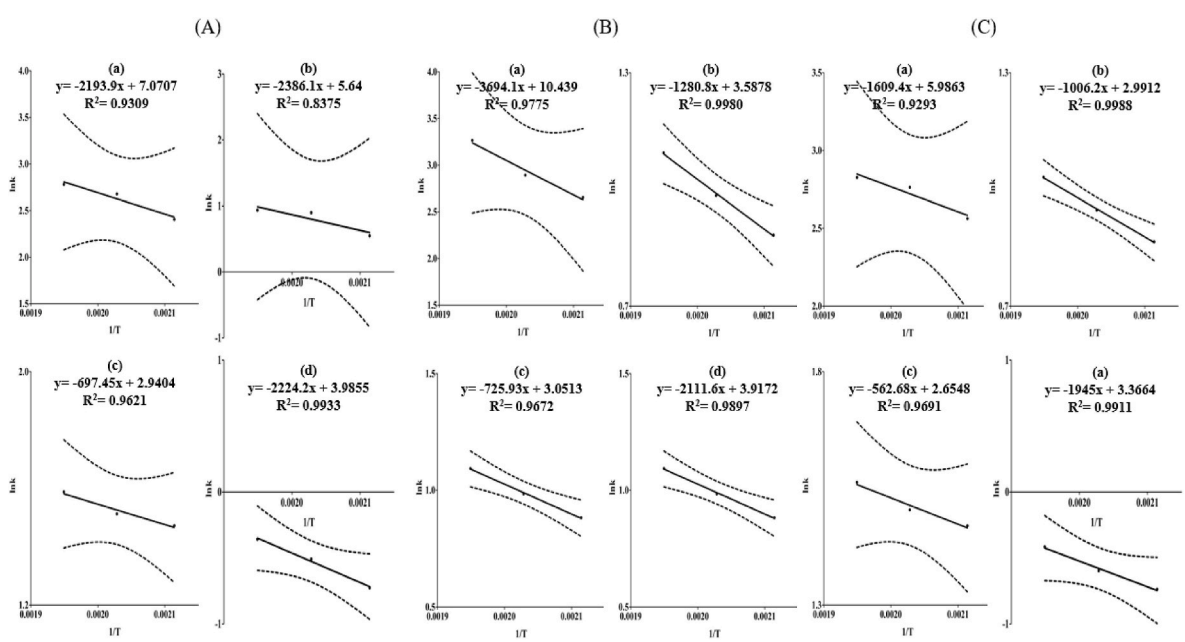

Fig. 1. Linear regression of Arrhenius equation in hot air (A), pan (B), and Air fryer (C) roasted coffee samples; (a) α-DCs formation, (b) α-DCs degradation, (c) melanoidins formation, (d) crude proteins degradation.

The E_a values for the melanoidin formation stages of the hot air, pan, and air fryer roasting methods were 5.80, 6.04, and 4.69 kJ/mol, respectively. In hot air roasting, the E_a value for the formation stage was 23.7% higher than that in air fryer roasting. Moreover, in pan roasting, the E_a value for the formation stage was 28.8% higher than that in air fryer roasting.

In the crude-protein degradation stage, the E_a values for the hot air, pan, and air fryer roasting methods were 18.49, 17.56, and 16.17 kJ/mol, respectively. The E_a value for the degradation stage of hot air roasting was 14.4% higher than that of air fryer roasting. The E_a value for the degradation stage of pan roasting was also 8.60% higher than that of air fryer roasting.

On comparing the three roasting methods, air fryer roasting evidently reduced the E_a values of α-DCs, melanoidins, and crude proteins by up to 57.8%, 22.3%, and 12.5%, respectively. A lower E_a value implies a smaller response to temperature changes (Cha & Lee, 2020; Yoo, Lee, Lee, & Lee, 2022). Therefore, among the three roasting methods, air fryer roasting is known to be the least sensitive to temperature changes.

3.7. Colour values

Colour values are expressed in terms of L*, a*, and b*, which represent lightness–darkness (L*), redness–greenness (a*), and yellowness–blueness (b*), respectively. The results are presented in Table 3. Previous studies have classified L* values into three roasting degrees: light (L* > 25), medium (L* 23–25), and dark (L* < 23) (Ludwig et al., 2013; Toci, Azevedo, & Farah, 2020). Among these, medium roast coffee has the highest concentration of most volatile compounds and is also the most preferred and consumed coffee roast worldwide (Toci et al., 2020). In this study, medium roast coffee is produced at temperatures of 240, 220, and 200 °C via hot air roasting for 12, 15, and 21 min; pan roasting for 6, 12, and 15 min; and air fryer roasting for 12, 15, and 18 min, respectively. On comparing the α-DC concentration results displayed in Table 1, the highest α-DC concentration is found in medium roasting, which is the most preferred roasting degree by consumers. Furthermore, on comparing roasting methods, pan roasting at 240 °C for 6 min evidently reduces α-DCs, toxic compounds, by up to 23.0 %.

4. Conclusion

This study reveals the changes in the concentration of Maillard reaction products, namely, α-DCs and melanoidins, and reactant crude protein in coffee samples prepared using various roasting methods. During the roasting process, α-DC concentration initially increases and subsequently decreases because it is an intermediate product, while melanoidin concentration increases as the Maillard reaction progresses. Crude protein, a Maillard reactant, decreases as the Maillard reaction progresses. In particular, kinetic modelling demonstrated that pan roasting produces and decomposes the above substances the fastest. The Arrhenius equation revealed that air fryer roasting has the lowest E_a value, confirming that it is the roasting method least affected by temperature changes. Colorimetric results confirmed that the highest α-DC concentration is yielded in medium roasting, which produces the most preferred coffee by consumers. In this study, toxic substances, such as α-DCs, were found at the medium roasting level. Among the three roasting methods, pan roasting exhibits the lowest α-DC content and the highest browning. This study provides the relevant with insight into the prediction and regulation of various compounds in roasted coffee beans based on various roasting methods.

CRediT authorship contribution statement

Seongho Lee: Writing – original draft, Methodology, Investigation, Formal analysis. **Eunyeong Choi:** Validation, Methodology. **Kwang-Geun Lee:** Writing – review & editing, Writing – original draft,

Table 3
Color values of coffee samples by various roasting.

Roasting method	Roasting temperature (°C)	Roasting time (min)	Color value		
			L*	a*	b*
Hot air	200	1	66.70 ± 0.01^{j}	2.50 ± 0.01^{a}	20.42 ± 0.01^{h}
		3	65.51 ± 0.01^{i}	3.65 ± 0.01^{b}	24.05 ± 0.02^{h}
		6	64.28 ± 0.03^{h}	7.78 ± 0.02^{c}	31.13 ± 0.02^{i}
		12	47.36 ± 0.02^{g}	14.74 ± 0.02^{h}	35.23 ± 0.04^{g}
		15	30.58 ± 0.01^{f}	15.58 ± 0.04^{i}	25.91 ± 0.02^{f}
		18	28.71 ± 0.02^{e}	14.80 ± 0.02^{h}	23.43 ± 0.01^{e}
		21	24.55 ± 0.04^{d}	12.94 ± 0.03^{g}	17.54 ± 0.03^{d}
		24	22.47 ± 0.02^{c}	12.10 ± 0.09^{f}	15.87 ± 0.05^{c}
		27	19.27 ± 0.05^{b}	10.43 ± 0.09^{e}	12.21 ± 0.11^{b}
		30	17.55 ± 0.04^{a}	7.95 ± 0.05^{d}	8.73 ± 0.05^{a}
	220	1	66.12 ± 0.00^{j}	2.84 ± 0.00^{a}	21.55 ± 0.00^{h}
		3	65.21 ± 0.00^{i}	3.52 ± 0.01^{b}	23.63 ± 0.01^{h}
		6	59.74 ± 0.00^{h}	10.83 ± 0.02^{g}	33.72 ± 0.01^{h}
		12	30.47 ± 0.00^{g}	15.26 ± 0.00^{j}	25.90 ± 0.04^{g}
		15	23.37 ± 0.01^{f}	12.76 ± 0.04^{i}	15.87 ± 0.05^{f}
		18	20.97 ± 0.12^{e}	11.47 ± 0.05^{h}	13.23 ± 0.05^{e}
Hot air	220	21	18.33 ± 0.12^{d}	9.80 ± 0.08^{f}	11.67 ± 0.12^{d}
		24	17.07 ± 0.05^{c}	8.57 ± 0.05^{e}	8.77 ± 0.09^{c}
		27	16.77 ± 0.09^{b}	7.07 ± 0.05^{d}	8.23 ± 0.05^{b}
		30	15.60 ± 0.08^{a}	6.43 ± 0.09^{c}	6.20 ± 0.08^{a}
	240	1	66.40 ± 0.00^{i}	3.18 ± 0.03^{a}	22.48 ± 0.00^{h}
		3	65.20 ± 0.00^{j}	3.75 ± 0.03^{b}	24.54 ± 0.00^{i}
		6	52.01 ± 0.00^{h}	13.50 ± 0.00^{j}	34.92 ± 0.00^{j}
		12	24.52 ± 0.01^{g}	12.74 ± 0.00^{i}	17.17 ± 0.04^{g}
		15	22.90 ± 0.08^{f}	9.97 ± 0.00^{h}	11.53 ± 0.05^{f}
		18	17.04 ± 0.00^{e}	8.33 ± 0.09^{g}	8.74 ± 0.05^{e}
		21	16.14 ± 0.03^{d}	6.91 ± 0.10^{f}	7.23 ± 0.09^{d}
		24	15.43 ± 0.12^{c}	6.09 ± 0.11^{e}	6.11 ± 0.21^{c}
		27	15.20 ± 0.13^{b}	5.79 ± 0.06^{d}	5.58 ± 0.10^{b}
		30	14.25 ± 0.30^{a}	4.99 ± 0.10^{c}	4.86 ± 0.20^{a}
Pan	200	1	65.66 ± 0.02^{i}	3.49 ± 0.00^{a}	23.96 ± 0.02^{j}
		3	63.24 ± 0.03^{j}	4.24 ± 0.03^{b}	25.50 ± 0.06^{i}
		6	50.35 ± 0.04^{h}	8.94 ± 0.04^{f}	28.87 ± 0.08^{h}
		12	31.22 ± 0.57^{g}	14.82 ± 0.16^{j}	23.36 ± 0.69^{g}
Pan	200	15	22.24 ± 0.07^{f}	12.31 ± 0.04^{i}	15.71 ± 0.03^{f}
		18	21.33 ± 0.02^{e}	11.54 ± 0.02^{h}	13.97 ± 0.01^{e}

(continued on next page)

Table 3 (*continued*)

Roasting method	Roasting temperature (°C)	Roasting time (min)	Color value		
			L*	a*	b*
		21	18.54 ± 0.02^{d}	9.27 ± 0.05^{g}	8.40 ± 0.02^{d}
		24	17.35 ± 0.03^{c}	7.96 ± 0.06^{e}	8.03 ± 0.01^{c}
		27	16.35 ± 0.02^{b}	6.94 ± 0.03^{d}	6.88 ± 0.03^{b}
		30	15.55 ± 0.04^{a}	6.29 ± 0.02^{c}	5.91 ± 0.05^{a}
	220	1	63.75 ± 0.01^{j}	3.96 ± 0.02^{a}	24.78 ± 0.01^{h}
		3	60.37 ± 0.02^{i}	5.91 ± 0.04^{c}	27.03 ± 0.05^{i}
		6	41.87 ± 0.40^{h}	15.53 ± 0.11^{i}	31.95 ± 0.46^{h}
		12	23.59 ± 0.39^{g}	12.06 ± 0.04^{h}	15.11 ± 0.13^{g}
		15	20.68 ± 0.08^{f}	10.64 ± 0.06^{g}	12.93 ± 0.01^{f}
		18	18.67 ± 0.03^{e}	8.82 ± 0.12^{f}	7.99 ± 0.17^{e}
		21	17.71 ± 0.03^{d}	7.56 ± 0.17^{e}	7.99 ± 0.17^{d}
		24	16.67 ± 0.06^{c}	6.62 ± 0.05^{d}	6.50 ± 0.11^{c}
		27	15.07 ± 0.04^{b}	5.91 ± 0.03^{c}	4.91 ± 0.05^{b}
	240	1	64.56 ± 0.11^{j}	4.26 ± 0.04^{b}	24.52 ± 0.10^{h}
		3	56.56 ± 0.06^{i}	9.95 ± 0.04^{g}	30.30 ± 0.22^{i}
		6	24.74 ± 0.02^{h}	13.63 ± 0.05^{h}	18.72 ± 0.06^{g}
Pan	200	12	16.80 ± 0.12^{g}	5.86 ± 0.07^{f}	5.93 ± 0.03^{f}
		15	15.55 ± 0.16^{f}	5.44 ± 0.20^{e}	4.76 ± 0.06^{e}
		18	14.74 ± 0.02^{e}	4.93 ± 0.03^{d}	4.56 ± 0.03^{d}
		21	13.21 ± 0.03^{d}	4.60 ± 0.07^{c}	4.39 ± 0.05^{d}
		24	11.60 ± 0.03^{c}	5.07 ± 0.01^{d}	4.12 ± 0.06^{c}
		27	11.30 ± 0.03^{b}	4.36 ± 0.04^{b}	3.29 ± 0.06^{b}
		30	10.97 ± 0.03^{a}	3.58 ± 0.15^{a}	2.76 ± 0.00^{a}
Air fryer	200	1	65.66 ± 0.02^{i}	3.49 ± 0.00^{a}	23.96 ± 0.02^{j}
		3	63.24 ± 0.03^{j}	4.24 ± 0.03^{b}	25.50 ± 0.06^{i}
		6	50.35 ± 0.04^{h}	8.94 ± 0.04^{f}	28.87 ± 0.08^{h}
		12	31.22 ± 0.57^{g}	14.82 ± 0.16^{j}	23.36 ± 0.69^{g}
		15	22.24 ± 0.07^{f}	12.31 ± 0.04^{i}	15.71 ± 0.03^{f}
		18	21.33 ± 0.02^{e}	11.54 ± 0.02^{h}	13.97 ± 0.01^{e}
		21	18.54 ± 0.02^{d}	9.27 ± 0.05^{g}	8.40 ± 0.02^{d}
		24	17.35 ± 0.03^{c}	7.96 ± 0.06^{e}	8.03 ± 0.01^{c}
		27	16.35 ± 0.02^{b}	6.94 ± 0.03^{d}	6.88 ± 0.03^{b}
		30	15.55 ± 0.04^{a}	6.29 ± 0.02^{c}	5.91 ± 0.05^{a}
Air fryer	220	1	65.62 ± 0.04^{i}	3.52 ± 0.01^{b}	23.63 ± 0.01^{j}
		3	66.12 ± 0.00^{j}	2.84 ± 0.00^{a}	21.55 ± 0.00^{i}
		6	59.74 ± 0.00^{h}	10.83 ± 0.02^{g}	33.72 ± 0.01^{h}
		12	31.87 ± 0.08^{g}	17.72 ± 0.09^{j}	26.85 ± 0.10^{g}
		15	22.21 ± 0.05^{f}	12.92 ± 0.07^{i}	16.19 ± 0.02^{f}
		18	20.67 ± 0.08^{e}	11.82 ± 0.10^{h}	15.87 ± 0.08^{e}
		21	19.64 ± 0.01^{d}	10.66 ± 0.09^{f}	12.46 ± 0.01^{d}
		24	16.45 ± 0.09^{c}	7.42 ± 0.02^{e}	8.59 ± 0.02^{c}
		27	15.71 ± 0.05^{b}	6.84 ± 0.02^{d}	6.56 ± 0.05^{b}
		30	14.73 ± 0.04^{a}	6.48 ± 0.04^{c}	6.17 ± 0.02^{a}
	240	1	64.56 ± 0.11^{j}	4.26 ± 0.04^{b}	24.52 ± 0.10^{h}
		3	56.56 ± 0.06^{i}	9.95 ± 0.04^{g}	30.30 ± 0.22^{i}
		6	24.74 ± 0.02^{h}	13.63 ± 0.05^{h}	18.72 ± 0.06^{g}
		12	16.80 ± 0.12^{g}	5.86 ± 0.07^{f}	5.93 ± 0.03^{f}
		15	15.55 ± 0.16^{f}	5.44 ± 0.20^{e}	4.76 ± 0.06^{e}
		18	14.74 ± 0.02^{e}	4.93 ± 0.03^{d}	4.56 ± 0.03^{d}
		21	13.21 ± 0.03^{d}	4.60 ± 0.07^{c}	4.39 ± 0.05^{d}
Air fryer roasting	200	24	11.60 ± 0.03^{c}	5.07 ± 0.01^{d}	4.12 ± 0.06^{c}
		27	11.30 ± 0.03^{b}	4.36 ± 0.04^{b}	3.29 ± 0.06^{b}
		30	10.97 ± 0.03^{a}	3.58 ± 0.15^{a}	2.76 ± 0.00^{a}

1) All values are shown as mean ± S.D. (standard deviation) (n = 3).
2) Lowercase letters (series "a-j") indicate significant ($p < 0.05$) differences in the same column.

Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.lwt.2024.116950.

Data availability

Data will be made available on request.

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